

Ước lượng runtime lượng tử

Bảng chứng từ QPU lưu trữ và simulator local

Phạm vi khoa học PARTIAL

Không chạy live QPU. Không gộp các clock khác nhau thành một bảng xếp hạng.

Dữ liệu và phạm vi

Miền	Đơn vị dữ liệu	Phạm vi
QPU Ma-Li	340 observations	Nguồn riêng
QPU Qonductor	4.482 observations	Nguồn riêng
QPU QPack	3.945 observations	46 workflows
Tổng QPU	8.767 observations	Labels archived
Simulator local	204 members / 191 hashes	162 core + 42 frontier

Đây là hai domain với target và clock riêng, không phải một leaderboard phẳng.

Thiết kế so sánh

Nguyên tắc	Áp dụng	Giới hạn
Rows chung	Graph V3-large và polynomial dùng cùng unified ledger và grouped outer folds	So sánh theo source-local shared rows
Fit	Preprocessing và calibration chỉ fit trên outer-train	Không chọn theo labels của test
Coverage	Unavailable và failures giữ trong denominator được giao	Coverage khác accuracy
Clock	Hiển thị output clock cùng evaluation target clock	Không xếp hạng chéo clock

MAE thấp hơn trên shared rows không tự động chứng minh superiority trên toàn ledger.

Coverage phương pháp

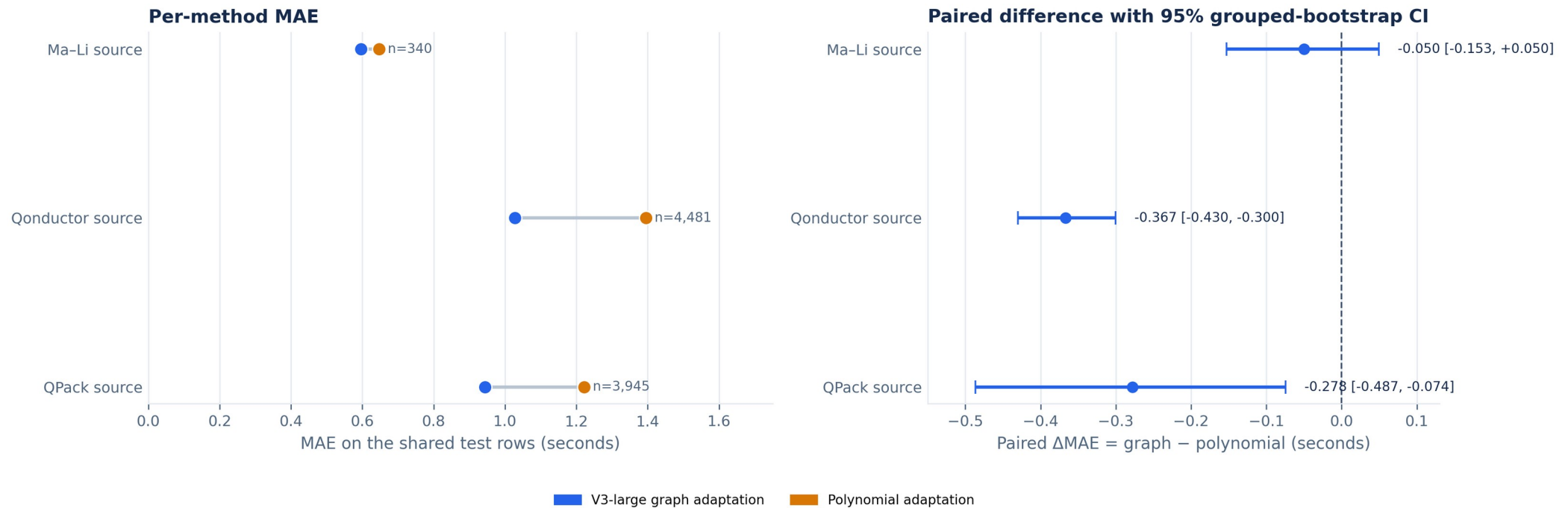
QPU approach	Evidence status	Simulator approach	Evidence status
Ma-Li graph	Unified adaptation; V3 and V3-large	Ma-Li local graph	162 core rows; supplementary diagnostic
Qonductor	Unified polynomial adaptation	Azizov	Classical adaptations only
Scholten	Nominal proxy; 5,769 / 8,767	Family-aware	Local joint task not evaluated
QCRE	Scheduled proxy; 8,766 / 8,767	Maestro	Pilot gate failed; no predictor score
Qiskit estimate_duration	Native schedule; 8,766 / 8,767	cuTensorNet	Same-plan estimate; 603 / 612 ok
Hyb-HANAS	One-circuit cost adaptation; 8,420 finite	Pasqal emu-MPS	Analog pilot not promoted

24 QPU variants are not 24 papers. Coverage is not accuracy; output clocks differ.

MAE theo source trên shared rows

Real-QPU source-local paired error

V3-large graph adaptation vs unified polynomial · lower MAE is better · same successful test IDs per source



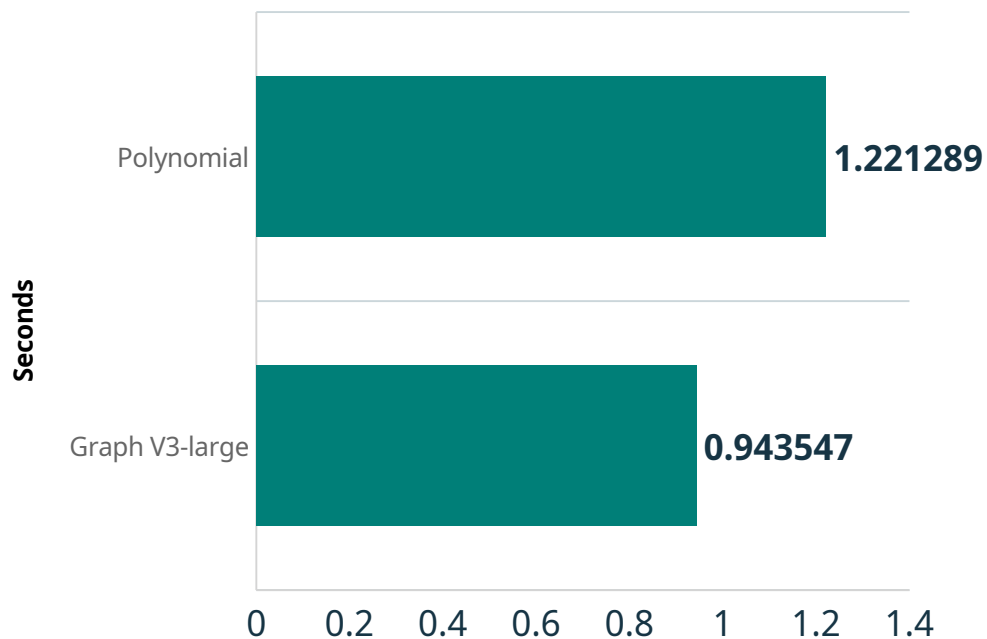
Source slices are descriptive, not separate source-trained models. Ma-Li CI crosses zero; Qonductor coverage is asymmetric (row4477 unavailable).

qpu/archived_method_pairwise_comparisons.csv SHA-256 46feff9632a48b7c852a938c0e5a153209d700e0574a5d8cd685c609ab5b1313

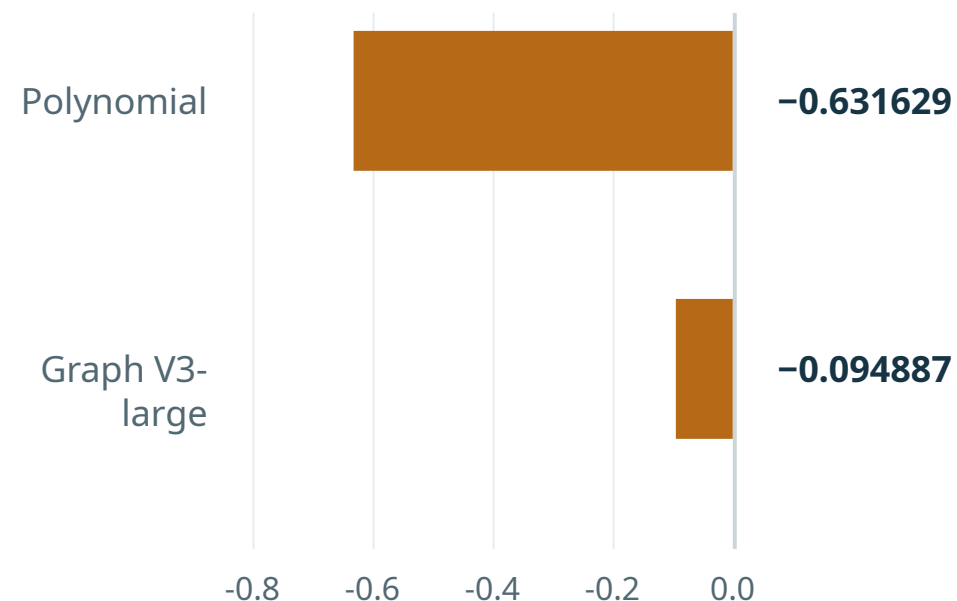
Shared rows by source. 95% grouped-bootstrap CI for graph minus polynomial MAE. Lower MAE favors the graph adaptation.

QPack: MAE và R^2

MAE (s)



R^2



Equal-workflow MAE diagnostic: 0.946290 s vs 1.217754 s across 46 workflows. Both R^2 values remain negative.

6 templates recur; workflow holdout is not unseen-template validation.

Scheduled duration và observed service time

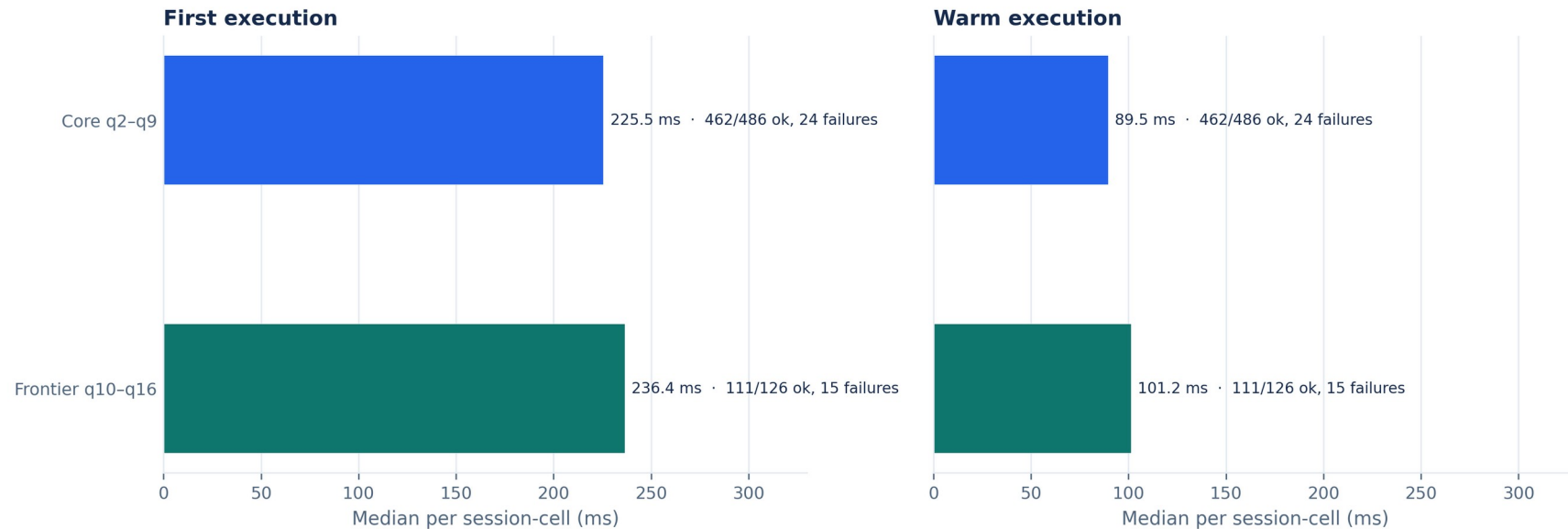
Route	Output clock	What it excludes or changes
QCRE critical path	Scheduled single-shot	No queue, service, transport, or fixed overhead
Qiskit estimate_duration	Scheduled single-shot	Native API output under frozen Target
Shot-scaled variants	Scheduled total proxy	Non-native multiplication by shots
Outer-train calibration	Calibrated service-time target	Adaptation, fit on outer-train only

8.766 predictions per raw route. Row4477 is unavailable. Scheduled duration does not replace observed QPU service time.

Simulator: coverage và quality gate

CUDA-Q MPS under a fixed quality gate

Same engine/configuration/precision; clocks and panel strata remain separate; failures stay in each denominator



Quality policy: fp64, bond 16, cutoff 1e-10, gesvdj, fidelity ≥ 0.99 . Medians summarize successful rows; failed cells remain in the denominators.

Descriptive within-engine/configuration comparison only; no cross-engine ranking or circuit-level paired claim.

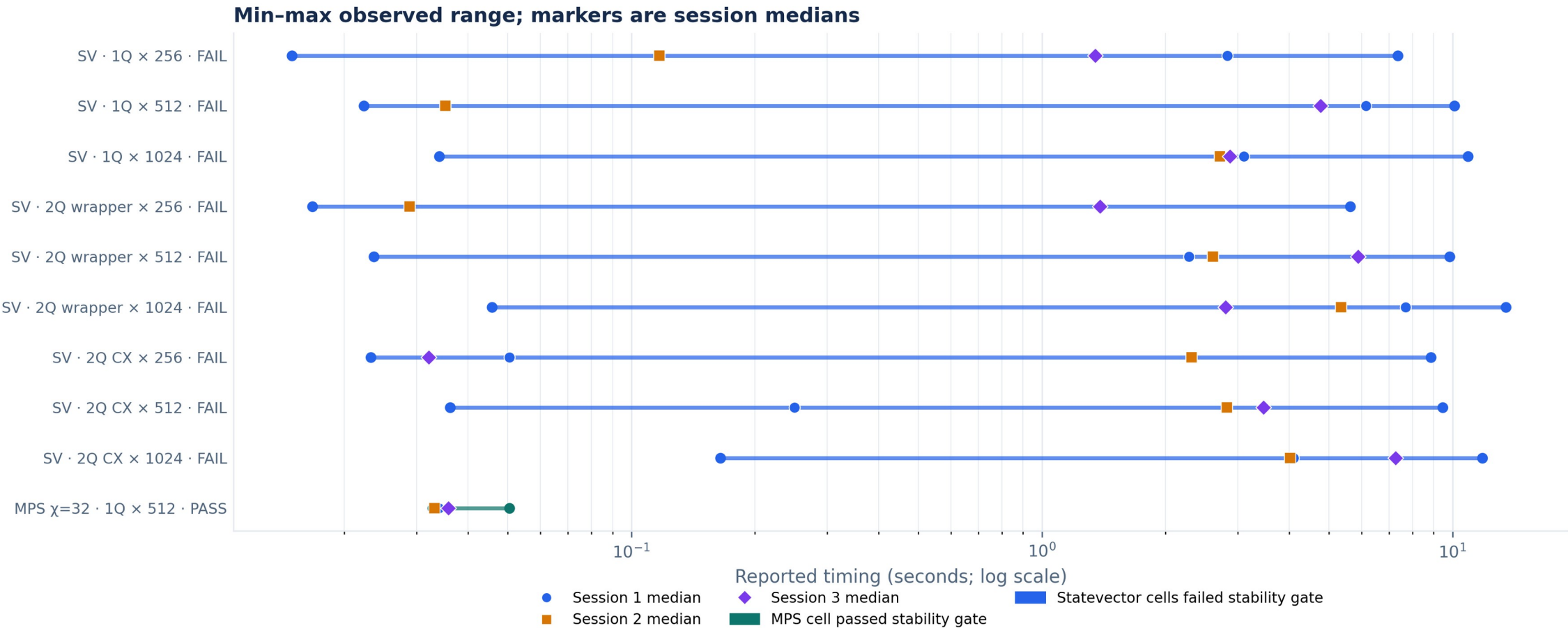
simulator/per_configuration_metrics.csv SHA-256 f8628e32fb0a0cecc8f06eb76a77b9e14e25b4ba8c288b2e72b8227d117d757a | simulator/availability_matrix.csv SHA-256 f3b48ecc2967d5bf8d2ce3b0368dc5e1c8523ceda5626070cde3979df1880458

MPS: 612 session-cells per clock, 573 ok, 21 quality_failed, 18 adapter_error. TN: 612 sessions, 603 ok, 9 timeout.

Panel 204 members / 191 exact hashes. Failures remain in coverage; session-cells are not distinct circuits.

Maestro pilot timing spread

10 synthetic calibration cells · process-isolated reported execution · timing stability gate failed overall



Terminal pilot: 9/9 statevector cells failed stability; 1/1 MPS cell passed. No accepted predictor or panel accuracy score; cause of spread is undetermined.

Kết luận và phạm vi còn thiếu

Có thể kết luận	Chưa đủ evidence để kết luận
Graph V3-large có MAE thấp hơn polynomial trên source-local shared rows.	Không có pooled cross-source superiority claim.
QPack có MAE thấp hơn nhưng cả hai R^2 đều âm.	Không nói model giải thích tốt biến thiên runtime.
Simulator coverage có quality/failure gates rõ ràng.	Run completed không đồng nghĩa accuracy benchmark hoàn tất.
Maestro pilot ghi nhận stability failure.	Không có accepted predictor hoặc panel accuracy score.

Scientific coverage: PARTIAL. Build validation không đồng nghĩa full replication hoặc public-release approval.

Fidelity phương pháp và tái dựng đầu vào

Mảng	Đã dùng	Giới hạn fidelity
QPU graph	Ma-Li/QPack logical; Qonductor exact submitted physical QASM	Không thực thi compiled-input promise cho Ma-Li/QPack
QPack replay	6 reconstructed structures; representative rz(0.3)/rx(0.2)	Không có original optimizer angles / submitted routing
Graph features	7 global features; DAG + global MLP branches	Polynomial có 5 features; không cô lập causal graph effect
Native estimates	Snapshot calibration theo backend	Không phải row-day historical calibration
Unmeasured routes	Family-aware, original Composer, joint quality	Chưa có evaluated evidence phù hợp

Archived labels vẫn là runtime quan sát; replay reconstruction không thay đổi labels.

Clocks, quality, và release boundary

Domain	Clock or outcome	Interpretation
QPU target	Archived observed service/execution	Evaluation target, not schedule duration
QPU analytical	Scheduled single-shot / shot-scaled	Proxy; provider overhead excluded
Simulator	First / warm execution; quality gate	Keep engine, config, and clock separate
cuTensorNet	Selected-plan estimate	Compare only with contraction under same plan
Maestro	Process-isolated reported execution	Fresh process, terminal pilot failure
Package status	Local review stage	Validation is not full replication; clean-clone pass not claimed
Public release	Rights decision open	No public-release approval from this deck

Unavailable rows, timeouts, adapter errors, and quality failures stay in coverage denominators.

Appendix: coverage của 24 QPU variants

Variant	Pred / 8,767	Cover	Output clock
MaLi graph V3	8,763	99.95%	Observed service
MaLi graph V3-large	8,766	99.99%	Observed service
Unified polynomial V3	8,767	100.00%	Observed service
QCRE CP raw	8,766	99.99%	Scheduled CP
QCRE CP affine	8,766	99.99%	Cal. service
QCRE CP log-affine	8,766	99.99%	Cal. service
QCRE shot raw	8,766	99.99%	Shot schedule
QCRE shot affine	8,766	99.99%	Cal. service
QCRE shot log-affine	8,766	99.99%	Cal. service
Qiskit duration raw	8,766	99.99%	Scheduled 1-shot
Qiskit duration affine	8,766	99.99%	Cal. service
Qiskit duration log-affine	8,766	99.99%	Cal. service

Variant	Pred / 8,767	Cover	Output clock
Qiskit shot raw	8,766	99.99%	Shot schedule
Qiskit shot affine	8,766	99.99%	Cal. service
Qiskit shot log-affine	8,766	99.99%	Cal. service
Hyb 1-circuit raw	8,420	96.04%	T_eff
Hyb 1-circuit affine	8,420	96.04%	Cal. service
Hyb 1-circuit log-affine	8,420	96.04%	Cal. service
Hyb shot-linear raw	8,420	96.04%	Shot-linear cost
Hyb shot-linear affine	8,420	96.04%	Cal. service
Hyb shot-linear log-affine	8,420	96.04%	Cal. service
Scholten nominal raw	5,769	65.80%	CLOPSv-like
Scholten affine	5,769	65.80%	Cal. service
Scholten log-affine	5,769	65.80%	Cal. service